ECE 329 **Homework 5 — Solutions**

1. (10 pts) Starting with the left-hand side of the vector identity given in the problem, we can write

$$\begin{aligned}\mathbf{H}\cdot\nabla\times\mathbf{E}-\mathbf{E}\cdot\nabla\times\mathbf{H} &= (4\cos(\beta z)\hat{y})\cdot\begin{vmatrix}\hat{x}&\hat{y}&\hat{z}\\0&0&\frac{\partial}{\partial z}\\2\cos(\beta z)&0&0\end{vmatrix}-(2\cos(\beta z)\hat{x})\cdot\begin{vmatrix}\hat{x}&\hat{y}&\hat{z}\\0&0&\frac{\partial}{\partial z}\\0&4\cos(\beta z)&0\end{vmatrix}\\ &= (4\cos(\beta z)\hat{y})\cdot(-2\sin(\beta z)\hat{y})-(2\cos(\beta z)\hat{x})\cdot(4\cos(\beta z)\hat{x})\\ &= -8\beta\sin(2\beta z).\end{aligned}$$

The right-hand side of the vector identity is solved by

$$\begin{aligned}\nabla\cdot(\mathbf{E}\times\mathbf{H}) &= \nabla\cdot(2\cos(\beta z)\hat{x}\times 4\cos(\beta z)\hat{y})\\ &= \nabla\cdot\left(8\cos^2(\beta z)\hat{z}\right)\\ &= \frac{\partial}{\partial z}\left(8\cos^2(\beta z)\right)=-8\beta\sin(2\beta z).\end{aligned}$$

Consequently, the vector identity is verified.

2.

a) (7 pts) Referring to the hint, we expect $\mathbf{E}$, $\mathbf{B}$, and $\mathbf{V}_s$ to be orthogonal to one another. Based on this assumption, we can write

$$|\mathbf{E}|=\frac{Q}{4\pi\epsilon_o r^2}$$

for the magnitude of the electric field, and

$$|\mathbf{B}|=\frac{\mu_o I|d\mathbf{l}|}{4\pi r^2}$$

where $\mathbf{I}\,|d\mathbf{l}|$ can be expressed as QV_s. Since we are asked to show that the magnetic force exposed on q is as strong as the electric force only in the limit when $\upsilon_q\cdot V_s\to c^2\equiv\frac{1}{\mu_o\epsilon_o}$, we should expect the magnetic force and electric force on the test charge to be equal and along the opposite directions. Hence, we write

$$\begin{aligned}q\frac{Q}{4\pi r^2}\frac{1}{\epsilon_o} &= q\upsilon_q\mu_o\frac{Q}{4\pi r^2}V_s\\ c^2\mu_o &= \frac{1}{c^2\epsilon_o}\upsilon_q V_s\\ c^2 &\approx \upsilon_q V_s,\end{aligned}$$

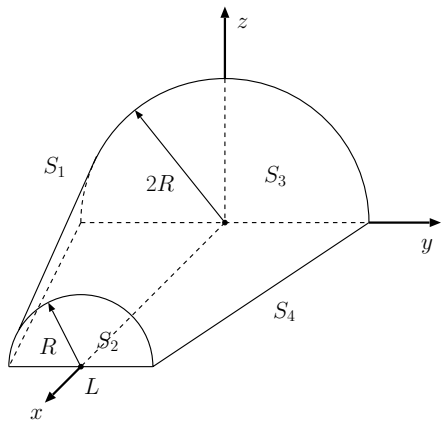
which shows the limit that the magnetic force can be as strong as electrig field on a moving test charge.

b) (3 pts) The field effects we observed in everyday life are collective interactions due to many charge sources . The electrostatic fields arising from elementary charges cancel each other statistically in most cases. While the magnetic fields arising from elementary charges in motion adds up to build appreciable macroscopic effects in cases with magnets.

3. Applying Gauss' law $\oint_S\mathbf{B}\cdot d\mathbf{S}=0$ and defining the closed surface S composed by the partial cone surface S_1, the semicircle S_2 with radius R, the semicircle S_3 with radius $2R$, and the trapezoid S_4

with bases $2R$ and $4R$ and height L as shown in the below figure, we can rewrite Gauss' law as, (2 pts)

$$\oint_S \mathbf{B}\cdot d\mathbf{S} = \int_{S_1} \mathbf{B}\cdot d\mathbf{S}_1 + \int_{S_2} \mathbf{B}\cdot d\mathbf{S}_2 + \int_{S_3} \mathbf{B}\cdot d\mathbf{S}_3 + \int_{S_4} \mathbf{B}\cdot d\mathbf{S_4} = 0.$$



Now, the flux through each surface can be calculated as follows. On the semicircle S_2, the magnetic flux is given by (2 pts)

$$\int_{S_2} \mathbf{B}\cdot d\mathbf{S}_2 = \int_{S_2} \left(a\hat{x} + a\hat{y} - a\frac{\pi R}{L}\hat{z}\right)\cdot \hat{x}\, dS_2 = a\cdot \text{Area}_2 = a\frac{\pi R^2}{2}.$$

On the semicircle S_3, the magnetic flux is given by (2 pts)

$$\int_{S_3} \mathbf{B}\cdot d\mathbf{S}_3 = \int_{S_3} \left(a\hat{x} + a\hat{y} - a\frac{\pi R}{L}\hat{z}\right)\cdot (-\hat{x})\, dS_3 = -a\cdot \text{Area}_3 = -a\frac{\pi\,(2R)^2}{2} = -2a\pi R^2.$$

On the trapezoid S_4, the magnetic flux is given by (2 pts)

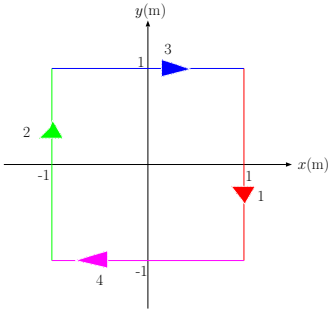
$$\int_{S_4} \mathbf{B}\cdot d\mathbf{S}_4 = \int_{S_4} \left(a\hat{x} + a\hat{y} - a\frac{\pi R}{L}\hat{z}\right)\cdot (-\hat{z})\, dS_4 = a\frac{\pi R}{L}\cdot \text{Area}_4 = a\frac{\pi R}{L}\left(\frac{2R+4R}{2}L\right) = 3a\pi R^2.$$

Finally, the magnetic flux through S_1 is calculated as (2 pts)

$$\begin{aligned}\int_{S_1} \mathbf{B}\cdot d\mathbf{S}_1 &= -\left(\int_{S_2} \mathbf{B}\cdot d\mathbf{S}_2 + \int_{S_3} \mathbf{B}\cdot d\mathbf{S}_3 + \int_{S_4} \mathbf{B}\cdot d\mathbf{S_4}\right)\\ &= -\left(\frac{a\pi R^2}{2} - 2a\pi R^2 + 3a\pi R^2\right) = -\frac{3a\pi R^2}{2}.\end{aligned}$$

4.

a) (1 pt) The 2-D sketch of the described configuration is shown below.



b)

i. (3 pts) Case : $|x| < 1$, and $|y| < 1$

The magnetic field distribution of the system of current sheets in the problem for part i is given in the .

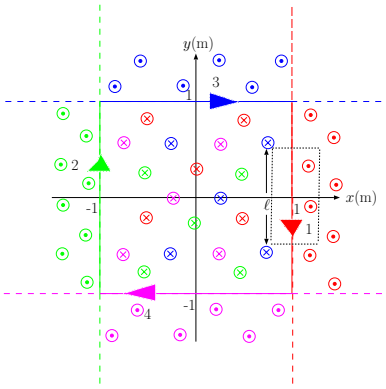


Figure 1: $|x| < 1$, and $|y| < 1$.

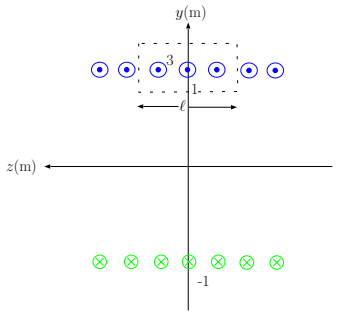
Using Ampere's law, $\nabla \times \mathbf{H} = \frac{\partial \mathbf{E}}{\partial t} + \mathbf{J}$ and Stoke's theorem, we can write

$$\oint_C \mathbf{H} \cdot d\ell = \int \mathbf{J}_s \cdot d\ell,$$

as $I = \int \mathbf{J} \cdot d\mathbf{S} = \int \mathbf{J}_s \cdot d\ell$. As seen in Figure 1, we can write

$$\mathbf{H}_3 = \begin{cases} -\hat{z}H_o & \text{for y<1,} \\ \hat{z}H_o & for\, y > 1, \end{cases} \tag{1}$$

for the current sheet located on $y = 1\,\text{m}$. We can calculate the path integral of $\mathbf{H}$ shown in in the following figure as $\oint_C \mathbf{H} \cdot d\ell = 2H_o\ell$, from which we find $H_o = \frac{J_s}{2}$.



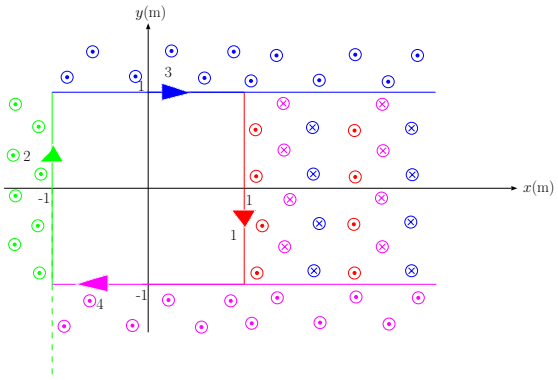
Then, we can write (1) in the following form

$$\mathbf{H}_3 = \begin{cases} -\hat{z}\frac{J_s}{2} & \text{for y<1,} \\ \hat{z}\frac{J_s}{2} & for\, y > 1. \end{cases}$$

The total $\mathbf{H}$ in the region between current sheets will be the superposition of $\mathbf{H}$ due to each current sheet $\mathbf{H} = \Sigma_i \mathbf{H}_i$, where $i = 1, 2, 3, 4$. Then, we find $\mathbf{H} = -\hat{z}2J_s = -\hat{z}4\,\frac{A}{m}$.

ii. (3 pts) Case : $x > 1$and $|y| < 1$

We can compute the region between $x > 1$and $|y| < 1$ (shown in the following figure) by referring to the boundary conditions, namely $\hat{n} \times (\mathbf{H}^+ - \mathbf{H}^-) = \mathbf{J}_s$.



Then, we can adapt the equation for the current sheet located on $y = 1$ in the following form

$$\hat{x} \times (\mathbf{H_1}^+ - \mathbf{H}_1^-) = -\hat{y}J_s,$$

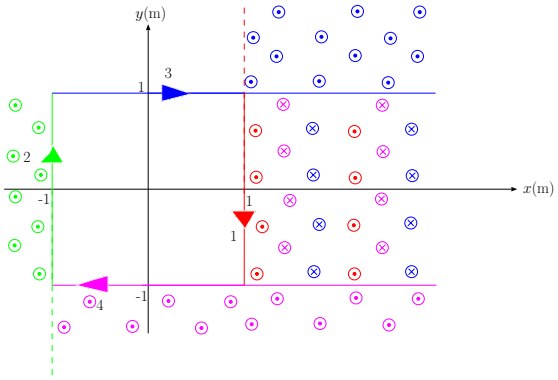
where $\mathbf{H}_1^-$ is the total $\mathbf{H}$ found in part i, and $\mathbf{H_1}^+ = \Sigma_i \mathbf{H}_i$,where $i = 1, 3, 4$. Thereby, we will write

$$\begin{aligned} \hat{x} \times (\mathbf{H_1}^+ - (-\hat{z}2J_s)) &= -\hat{y}J_s, \\ \hat{x} \times \mathbf{H_1}^+ - \hat{y}2J_s &= -\hat{y}J_s \\ \hat{x} \times \mathbf{H_1}^+ &= -\hat{y}J_s, \end{aligned}$$

from which we find $\mathbf{H}(x > 1, -1 < y < 1) = -\hat{z}J_s = -\hat{z}2\,\frac{A}{m}$.

iii. (3 pts) Case : $x > 1$, and $y > 1$

Once again, we will use the boundary conditions in order to compute the total $\mathbf{H}$ in the region of interest (shown in the following figure).



Boundary conditions along the current sheet located on $y = 1\,\text{m}$ will be

$$\hat{y} \times (\mathbf{H_3}^+ - \mathbf{H}_3^-) = \hat{x} J_s,$$

where $\mathbf{H}_3^-$ is the total $\mathbf{H}$ computed in part ii for $x > 1, -1 < y < 1$. Therefore, we write

$$\begin{aligned} \hat{y} \times (\mathbf{H_3}^+ - (-\hat{z} J_s)) &= \hat{x} J_s, \\ \hat{y} \times \mathbf{H_3}^+ + \hat{x} J_s &= \hat{x} J_s, \\ \hat{y} \times \mathbf{H_3}^+ &= 0, \end{aligned}$$

from which we find $\mathbf{H}(x > 1, y > 1) = 0$.

5.

a) (2 pts) Biot-Savart law tells us that the direction of the magnetic field generated by an infinitesimal current element is parallel to the cross product between the direction of the current and the vector joining the current element and the point under consideration. In our case, the current is parallel to $\hat{x}$. Then, using the right-hand rule, we can easily verify that, above the slab ($y > 0$), $\mathbf{B}$ should be parallel to $\hat{z}$,

$$\mathbf{B} \parallel 2\hat{x} \times \hat{y} \implies \mathbf{B} \parallel \hat{z},$$

while, below the slab ($y < 0$), $\mathbf{B}$ should be parallel to $-\hat{z}$,

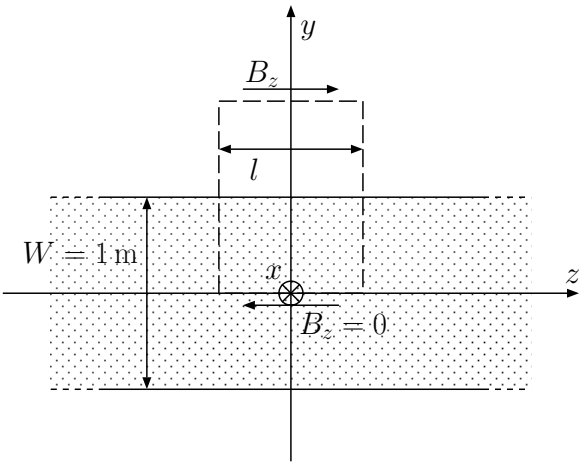
$$\mathbf{B} \parallel 2\hat{x} \times -\hat{y} \implies \mathbf{B} \parallel -\hat{z}.$$

Since the current distribution is symmetric with respect to the plane at $y = 0$, the fields at the same distance from above and below the slab should have equal amplitudes.

b) (2 pts) The magnetic fields slightly above and below $y = 0$ should have equal amplitudes but point in opposite directions. Exactly at $y = 0$ the fields must cancel each other such that

$$\mathbf{B}(y = 0) = \mathbf{0}\,\text{T}.$$

c) (2 pts) To compute $\mathbf{B}$ above the slab using Ampere's law, let us first define a rectangle R placed on the x-y plane as it is shown in the figure below.



Integrating $\mathbf{B}$ along the perimeter of the rectangle and noting that only the segment above the slab contributes to the integration, we have

$$\frac{1}{\mu_o}\oint_R \mathbf{B}\cdot d\mathbf{l} = \int_S \mathbf{J}\cdot d\mathbf{S}$$
$$\frac{1}{\mu_o}(B_z \times l) = 2 \times \frac{W}{2} \times l$$
$$B_z = \mu_o\,\mathrm{T}.$$

Similarly, we can compute $\mathbf{B}$ below the slab and obtain

$$\mathbf{B} = \begin{cases} \mu_o\,\hat{z}\,\mathrm{T} & \text{for } y > 0.5\,\mathrm{m}, \\ -\mu_o\,\hat{z}\,\mathrm{T} & \text{for } y < -0.5\,\mathrm{m}. \end{cases}$$

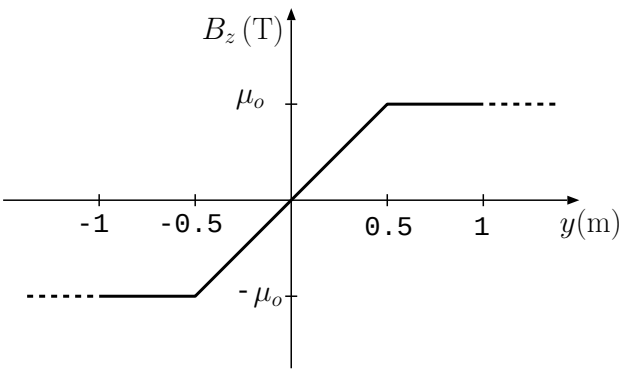
d) (2 pts) Applying the same technique, but considering a rectangle of height shorter than $W/2$ (0.5 m), we can compute $\mathbf{B}$ inside the slab (i.e., for $0 \le |y| \le 0.5\,\mathrm{m}$) as follows

$$\frac{1}{\mu_o}\oint_R \mathbf{B}\cdot d\mathbf{l} = \int_S \mathbf{J}\cdot d\mathbf{S}$$
$$\frac{1}{\mu_o}(B_z l) = 2yl$$
$$B_z = 2\mu_o y.$$

Finally, extending this result to negative z, we obtain

$$\mathbf{B} = 2\mu_o y\,\hat{z}\,\mathrm{T} \quad \text{for } -0.5\,\mathrm{m} \le y \le 0.5\,\mathrm{m}.$$

e) (2 pts) Plotting B_z as function of y.


B_z (T)
μ_o
-1
-0.5
0.5
1
y(m)
- μ_o